



Utilizing Downtime Analysis to Improve Availability

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

- Increasing equipment availability is the objective of all maintenance organizations. A robust and fact-based process to identify and address the reasons of unavailability is the most effective way to manage and sustainably increase availability.
- Some customers perform this analysis internally but others lack the resources or knowledge and Site Performance Managers from the Dealer and/or Caterpillar can support this process, creating a unique relationship with the customers' maintenance organization.
- This best practice describes methods to codify and analyze downtime, the process to analyze, how to identify opportunities, how to structure a customer meeting to review, and how to track action items.

2.0 Best Practice Description

2.1 Source of Data

- The source of downtime information can be dispatch data, maintenance records, or a combination of both.
 - Ideally, dispatch and Maintenance & Repair records are reconciled on a regular basis, therefore downtime categories are accurate in both systems.
- The quality & accuracy of the data used can substantially influence the analysis results. Measuring with sufficient granularity (detail) is important for the data to be meaningful.
- Quality of initial source data should not delay beginning the analysis because observations and improvements are possible with even poor-quality information.
- Processing data can be done in excel since most or all computerized maintenance management software (CMMS) have export functionality. Some dispatch or maintenance software programs have pre-programmed functionalities.

2.2 Time Usage Model

To successfully calculate the top tier and M&R process metrics, it is imperative that the mine site departments (production operations and maintenance) have a common understanding of the time usage model. Teams need to be aligned to ensure a common understanding of categories, terms, and calculations.

Caterpillar suggests the “time usage model” as shown in the table below. Customers may have different definitions or variations of the time usage model. When Caterpillar and dealers work with any site, it is advisable to ensure the time usage model and coding system are clearly understood.

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Total Calendar Hours						
Scheduled Hours						Unscheduled Hours (out of plan)
Available Hours (mechanically available; operational)				Downtime Hours Scheduled & Unscheduled Maintenance & Repairs (mechanically unavailable, not operational)		
Operating Hours (machine working)	Delays		Stand-by Hours Code List 3	Maintenance & Repairs	Delays	
	Operational Delay Hours Code List 1	Production Delay Hours Code List 2			Repair Delay Hours Code List 4	

Reference publication **PEBJ0190** Caterpillar Mining Equipment Management (MEM) Performance Metrics

2.3 Codification of Downtime

Each time a machine stops for maintenance or repair, a few critical things must be captured. It's preferable to capture any delay time, if applicable. Critical information for each stoppage should include:

- Stoppage type: Scheduled or Unscheduled
- Cause of the stoppage: PM, PCR, etc.
- Responsibility code: Maintenance, Operations, Other, etc.
- Responsible party: Customer, Dealer, Tire contractor, Fire suppression contractor, etc.
- Machine system or subsystem affected (engine, final drives, steering, etc.).

Repair delays cause the repair time to increase beyond the mine standards. Examples of delays might be due to "waiting for parts", "waiting for labor", or "waiting for facilities", etc. There are two primary methods to capture repair delays and either can assist the site to define continuous improvement opportunities.

- Capture all downtime at the machine system or subsystem level then record additional second level detail that defines the repair delays.
- Capture delay codes such as "waiting for parts", "waiting for labor", "waiting for facilities", etc. as primary downtime categories.

In this Best Practice example, the mine site uses AMT software, but similar structure can be used for any maintenance database tool (SAP, Wenco, JD Edwards). Fields of the database are presented in an Access table below.

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- For analysis reasons, it's recommended to use standard (predefined) downtime cause descriptions. This prevents duplication or inaccuracies from reasons such as spelling mistakes, difference in descriptions, etc.
- Caterpillar Service Management Control System (SMCS) codes are standardized information codes and are strongly recommended, especially where dealer personnel perform maintenance. [SMCS website](#)
 - The high-level structure for component codes includes a limited number of "basic" categories (less than 10). These include: engine (1000), transmission & drive line (3000), drive train (4000), implement controls (5000), machine (7000), etc.
 - Capturing these categories lacks sufficient granularity to action.
 - The detailed structure for component codes consists of a numeric table that offers significantly more codes (a few thousand). These include examples from the engine system such as: turbocharger (1052), camshaft (1210), piston (1214), etc.
 - Capturing at this level can be time consuming but can provide the additional level of detail that is desirable.
- Some mines will select a shorter list than the detailed structure noted above. Careful consideration should be given to decide the level of detail expected. It is advisable that this list should not be shorter than 100 codes.

COMPARTIMENT		JOB TYPE	
SMCS_CODE	SMCS_NAME	SMCS_CODE	SMCS_NAME
1363	AFTERCOOLER WATER PUMP	546	IDENTIFY
7279	AFTERTREATMENT FRAME	040	INSPECT
108U	AFTERTREATMENT SYSTEM	042	INSPECT & MAINTAIN
5510	AIR ACTUATOR	012	INSTALL
5504	AIR CHECK VALVE	089	INTERVIEW CUSTOMER ABOUT
1051	AIR CLEANER	549	KNURL
1803	AIR COMPRESSOR	500	LABOR EXCEEDING TRG
5502	AIR COMPRESSOR AIR LINE	508	LIFT & BLOCK
1804	AIR COMPRESSOR GOVERNOR	009	LIFT, BLOCK & LOWER
7320	AIR CONDITIONER	556	LOAD
1807	AIR CONDITIONER COMPRESSOR DRIVE	033	LOAD BANK TEST
1808	AIR CONDITIONER LINES	055	LOAD/UNLOAD
7322	AIR CONDITIONER RECEIVER/DRYER	526	LOOSEN
7337	AIR CONDITIONER, HEATER & PRESSURIZER CONTROL	525	LOOSEN & TIGHTEN
79N0	AIR CONTROL CYLINDER	509	LOWER
5500	AIR CONTROL SYSTEM	086	LUBRICATE
108A	AIR CONTROL VALVE	063	MACHINE/GRIND/MILL
5506	AIR CONTROL VALVE PNEUMATIC	082	MEASURE
4285	AIR DRYER	581	MEGATEST
79N1	AIR END COMPRESSOR	065	METALLIZE
1054	AIR FILTER ELEMENT	590	MODIFY
7452	AIR FILTER INDICATOR	551	MOVE
5508	AIR FILTER REGULATOR	554	OBTAIN
5509	AIR HORN CONTROL VALVE	572	OPEN
1050	AIR INDUCTION & EXHAUST SYSTEM	027	ORIGINAL INSTALLATION
0702	AIR INDUCTION & EXHAUST SYSTEM TOOL	520	OVERHAUL
1087	AIR INLET	075	PACKAGE
1090	AIR INLET HEATER	072	PAINT

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Field Name	Data Type	Description (Optional)
ID	AutoNumber	
Start	Date/Time	Start of DT event
Stop	Date/Time	Stop of DT event
Duration	Number	Duration of DT event in minutes
Equipment type	Short Text	example: shovel, excavator, dozer, truck
Fleet number	Short Text	example: EX-1, DT-2
Equipment model	Short Text	example: D9R, 777G, 6018
Planned	Yes/No	Can be planned or unplanned
Task	Short Text	Can be: PCR, PM or Repair
Variation code	Short Text	Can be: Waiting Labour, Waiting Lowboy (Transport), Waiting Machine etc
Variation duration	Number	Duration of Variation in minutes
Component	Short Text	Can be: engine, transmission, etc
Job type	Short Text	Can be: Adjust, weld, replace, etc

2.3.1 Data Structure

- Some of the fields described can be seen in this example below.
- Please note that the same maintenance stoppage (turbo leak) has two downtime entries, one waiting for parts and one for the actual repair.
- It should be emphasized that this is only (1) shutdown, otherwise this would impact MTBS and MTBF calculations & results.

Type of equipment	Equipment	Downtime cause	Downtime description	planned/ unplanned	Idle start	Idle end	Downtime (hr)
Bulldozer	Caterpillar D9R BD-01	250 Hr Service	250 Hr Service	Scheduled	29 Apr 2016 12:11:00	29 Apr 2016 16:31:00	4.33
Bulldozer	Caterpillar D9R BD-01	Daily service		Scheduled	11 Apr 2016 21:28:00	11 Apr 2016 21:39:00	0.18
Bulldozer	Caterpillar D9R BD-01	turbo leak	waiting for parts	Unscheduled	12 Apr 2016 17:30:00	13 Apr 2016 17:30:00	24.00
Bulldozer	Caterpillar D9R BD-01	turbo leak	Repair	Unscheduled	13 Apr 2016 17:30:00	13 Apr 2016 18:00:00	0.50

In order for the data to be easily analyzed, all fields of the table need to be codified. This is most efficiently performed with preselected categories or systems. SMCS codes can be used for Component description, but due to large number of options (~3,000) most mines use a shorter list of Component codes (see comments in the section “codification of downtime” above).

2.3.2 Non-Mechanical Causes and Delays

- It is important to capture repair delays because defining these delays provides opportunities to target areas for improvement (possible inputs to the Continuous Improvement (CI) process).
 - Measuring & reporting delay categories (“waiting for xx”) enables the team to define how much “pain” was caused and help motivate changes or improvements in the M&R organization (such as additional people, more or different types of tools, more parts, etc.).
- A percentage of the equipment downtime is not due to the repair itself, but delays due to waiting for parts, labor, tooling or other resources.
- A downtime analysis which does not include delays is incomplete and can lead to false conclusions.
- Waiting for parts, waiting for labor, etc. can be documented either in a separate entry, or as a variation on the repair (% of time, or hours).

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Table 1: List of non-mechanical delay codes

Attached below are reference tables extracted from the document titled “*Caterpillar Mining Equipment Management (MEM) Performance Metrics*”. It should be emphasized that different customers will have slightly different definitions and codes.

Code List 1	Production Delays
1A	Waiting For Equipment
1B	GET Change Out
1C	Blasting
1D	Tramming Equipment
1E	Equipment Diagnosis
1F	Engineering Delays
1G	Weather (Temporary)
1H	Other

Code List 2	Operational Delays
2A	Shift Change
2B	Change of Operator
2C	Pre-Shift Inspection
2D	Meeting
2E	Refuel/ Lube Service
2F	Meal Break
2G	Toilet Break
2H	Prayers
2I	Other

Code List 3	Stand-By/ Idle Time
3A	Not Required (Over Trucked)
3B	No Operator
3C	Maintenance Complete (On Ready Line)
3D	Equipment Retired
3E	Other

Out of Plan
Holiday
Accident
Weather
Strike/ Industrial Dispute
Act of God
Other

2.3.3 Handling Accidents and Equipment Abuse

- Sometimes the root cause of a component and machine failure is due to operational abuse.
- A downtime description, such as “accident”, or “operational abuse” needs to be utilized to differentiate those and allocate to the appropriate responsibility within the organization.

2.3.4 Handling Simultaneous Work and Backlog Performed at PM

- When jobs (work orders) are performed in parallel, downtime should be measured once. However, it is still expected that labor hours are accurately captured for each work order in the work order management system.
- Backlogs performed during PM within the PM “window of opportunity” or within a single shift can be allocated to PM downtime. Alternatively, downtime due to PM and downtime due to backlog repairs can be captured separately and allocated to the appropriate machine system.
 - It should be reinforced that assigning backlogs within the PM window of opportunity is acceptable only if the PM time is reasonable. Typically, most PM stoppages are targeted for completion within a single shift of slightly longer.

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- Note: A PM time of 47 hours is not reasonable. Therefore, it is suggested that downtime should be correctly and reasonably allocated to “PM time” and downtime allocated to backlogs should be correctly captured and allocated to that machine system (as an example frame welding).
- Backlogs performed at PM that cause the PM stoppage time to extend, need to be documented and allocated to the correct machine system / subsystem.
- However, a work order is still required for each repair and should include both system category and labor hours.

2.3.5 Handling Downtime from PM Execution

- Measure PM execution and compare with site goals and industry benchmarks.
- Often backlogs or repairs downtime is incorrectly allocated to PM, causing the downtime due to “PM” to be overinflated and downtime due to other reasons to be underreported in other machine systems (such as hydraulics, electrical, etc.).
 - Attention is needed to ensure this doesn’t drive the wrong conclusions.
 - As an example, if the top downtime drivers and impact on availability are used to prioritize improvement actions, then overreporting PM and underreporting other work could drive the wrong behavior.
- Trending of “PM unavailability” for trucks is typically more stable and consistent over time due to the size of fleets compared to fleets of a single unit or small fleets.
- When the 8-Step PM process is used, PM trending can be used by grouping all PM types together (PM1, PM2, PM3, PM4, PM5, etc.) for each machine-equipment model.
- If the 8-Step PM process is not used, then each machine model and PM type should be separated and reported as shown in the table below (i.e. PM250, PM500, PM1000, etc.).

Year	2015				
Month	Jul				
Machine Models	(All)				
Serial #	(All)				
Product Family/PM Interval	Total # of Machines Serviced	Average of Standard Hrs for PM	Average of Actual Hrs for PM	Total PM Hrs	Sum of Other Hrs
777D					
250	30	6.00	6.87	206.02	25.45
500	25	8.00	9.13	228.26	32.77
2000	7	12.00	14.39	100.71	18.50
1000	5	10.00	8.57	42.83	22.65
D9R					
250	15	6.00	5.70	85.46	
500	6	8.00	7.20	43.20	27.65
2000	4	12.00	10.41	41.64	17.70
1000	2	10.00	9.93	19.86	
992K					
250	11	6.00	7.40	81.39	45.22
500	4	8.00	9.90	39.59	
2000	1	12.00	9.50	9.50	10.70
1000	4	10.00	9.87	39.46	30.83

2.4 Processing Data and Analysis

- The information needs to be analyzed and displayed in a convenient format to allow the maintenance organization to investigate and determine opportunities to improve. Pareto charts of top 5 or top 10 are recommended. The impact of each DT category needs to be clearly indicated.
- It’s recommended that (2) types of Pareto analyses be reported for each machine fleet.
 - Top10 by downtime hours

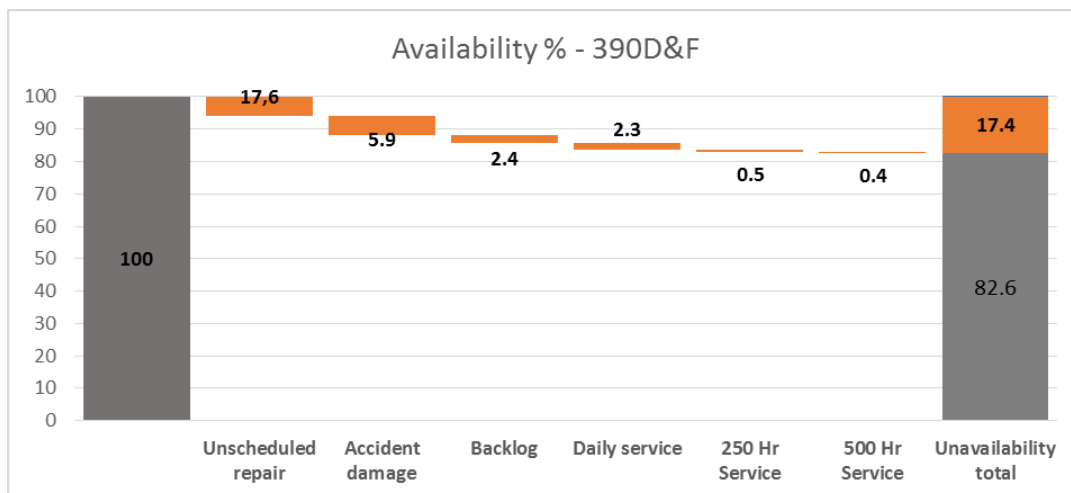
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- Top10 by stoppage events
- Processing data can be done in excel, since all software have the export functionality, but some dispatch or maintenance software have pre-programmed functionalities.

2.4.1 Identifying Top Issues / Significant Event Failures

- Pivot tables can be utilized to extract top downtime causes, by system and by delay code.
- Identify top 10 issues. Those should raise flags in investigating the root cause and the remedy.
- Compare downtime per category with industry benchmarks to identify gaps and potential opportunities (refer to metrics document or known benchmarks for that machine).

Cat code Equivalent	Unavailability (HRS)	Occurrences	Unavailability % (Impact on Availability)
PM	90.7	5	6.3%
Engine (CC)	59.3	1	4.1%
Daily Inspection	26.8	62	1.9%
GET	18.2	12	1.3%
Electrical System	17.5	16	1.2%
Swing System	14.3	10	1.0%
Engine	12.8	3	0.9%
Hydraulic System Distribution (Hoses, Lines, Tanks, Coolers)	11.2	5	0.8%
Frame/Chassis/Structures	9.2	1	0.6%
Bucket/Dipper	8.3	10	0.6%
Undercarriage	7.6	4	0.5%
Lubrication/Auto lube system	4.1	9	0.3%
Filters	3.1	6	0.2%
Accident Damage	2.7	1	0.2%
Cooling System	1.9	1	0.1%
Hydraulic System (Pumps & Motors)	1.8	1	0.1%
Grand Total	289	94	20%



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Attached below is a reference table for large mining trucks indicating reference information by machine system for MTBS, MTTR and availability impact. This table can be located in the "Caterpillar Mining Equipment Management (MEM) Performance Metrics" document publication **PEBJ0190**.

Generic Pareto Reference Large OHT's (785-793)			
Component / System	ref. MTBS (hours)	ref. MTTR (hours)	ref. Availability Impact (%)
Accidents	6750	28.00	0.37%
Air System & Starting	2100	2.80	0.12%
Air Conditioning	6650	0.30	0.00%
Auto Lube System	6100	0.90	0.01%
Base Machine	4000	0.30	0.01%
Brakes & Brk System	1200	6.00	0.45%
Cab / Operator Station	450	2.60	0.52%
Chassis	2850	7.00	0.22%
Cooling System	1300	5.10	0.35%
Differential	6950	12.90	0.17%
Dispatch System	800	0.60	0.07%
Dump Body	3750	6.90	0.16%
Electronics & Electrical	1250	1.90	0.14%
Engine	550	9.90	1.61%
Filters	2800	1.30	0.04%
Final Drives	2100	12.20	0.52%
Front Wheels	6150	5.30	0.08%
Hoist System	1350	3.10	0.21%
Hydraulic System	1250	4.90	0.35%
Mirrors	850	0.20	0.02%
miscellaneous	750	2.70	0.32%
PM	250	7.70	2.76%
Steering System	1950	6.20	0.29%
Suspension	3150	9.40	0.27%
Switches & Sensors	1950	1.90	0.09%
Tires & Rims	550	3.20	0.52%
Torque Converter	7250	5.10	0.06%
Transmission	900	5.30	0.53%
unknown	5850	6.20	0.10%

"BASE MACHINE":
Includes ladders, hand rails, sheet metal, brackets and other miscellaneous areas of the machine not covered by one of the description codes in the list.

CAB / OPERATOR STATION:
This description code includes cab, windshield, seat, windows, etc.

ELECTRONICS & ELECTRICAL:
Includes electrical / electronic components & systems, e.g. lighting, VIMS, wiring harnesses, connectors, etc.

ENGINE:
Includes all engine and engine-related components and systems not covered by another description code, e.g. basic engine, air intake & exhaust, fuel system, etc.

HYDRAULIC SYSTEM:
This description code includes all pumps, hydraulic actuators, hoses, lines, oil coolers, tanks, etc not included in another description code.

"MISCELLANEOUS":
Includes any shutdown event that is not covered by one of the other description codes.

NOTE: MTTR = 16.80 hours for a 500 hour PM interval. In other words, if MTBS for PM = 500 hours, MTTR = 16.80 hours.

"UNKNOWN":
Stoppage is unidentified or data is unavailable. An excessive number of unknown events is an indication of inadequate recordskeeping practices.

NOTE: If the PM interval is 500 hours, $MTBS_{PM} = 500$ hours, $MTTR_{PM} = 16.80$ hours and Impact of PM on Availability = 3.00%.

2.5 Identifying & Addressing Root Causes

- Technical reports and feedback from customer and dealer personnel will be required to understand and evaluate root cause of failures and other stoppages.
- At a high-level view, failures might be attributable to maintenance performance, operations conditions or behaviors, manufacturer machine or parts quality, or combination these.
- Continuous Improvement projects are often valuable opportunities to identify and address root cause of downtime.

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- Focus on cases or failures which may require contingency plans to protect the fleet, if issues can affect more units (parts, labor).

2.6 Availability and Availability Index

Caterpillar recommends using Availability index with physical availability.

- Physical Availability = (Calendar Hours - Downtime Hours) / Calendar Hours.
- Availability index = MTBS / (MTBS + MTTR)
- MTBS = Operating Hours / Number of Shutdowns
- MTTR = Downtime Hours / Number of Shutdowns

Low utilization (i.e., high standby hours) will artificially inflate Physical Availability (PA) and can provide the maintenance department with a false sense of "good performance".

The table below is an example that demonstrates this concept. In this example, 3 trucks have completely different reliability (MTBS), but because of differences in utilization, the physical availability results are the same. Therefore, the conclusion is that physical availability without understanding utilization can be deceiving. Availability index is a reporting metric that removes the differences due to changes in utilization. Availability index is a more objective way to measure the efficiency (speed) and effectiveness (quality) of the maintenance organization.

	Calendar Hours	Operating Hours	Downtime Hours	Number of Shutdowns	Available Hours	Utilization (%)	Availability (Physical)	MTBS (Hours)	MTTR (Hours)	Availability (Index)
Truck #1	2160	1500	300	20	1860	81%	86.1%	75	15	83.3%
Truck #2	2160	1000	300	20	1860	54%	86.1%	50	15	76.9%
Truck #3	2160	500	300	20	1860	27%	86.1%	25	15	62.5%

2.7 Follow up and Tracking Actions

An action plan is a critical part of implementing improvements. A 3W (Who, What, When) type management methodology is the recommended approach to improving availability.

It's recommended to conduct monthly meetings with the customer, dealer and Cat representative if possible. This team should include key members of the maintenance organization such as: site maintenance manager, senior planner, foreman, condition monitoring, reliability engineering, continuous improvement, and the Cat & dealer site performance managers.

In the below example a column is added every month to document progress towards each objective.

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MINING DEALER BEST PRACTICE

internal	machine model	who	what	when	start	status	update	update
yes	6020B	Key Account manager/ Site Manager	significant electrical downtime. consider proposing an electrician for 2 rotations of troubleshooting and hands on training to reduce downtime	15-Dec	08-Nov	ongoing	10 Aug: HMS expert will do 2 rotations, customer has hired two electrician experts	
no	6040FS	Logistics department	provide a quotation and lead time for 6040FS stick	30-Aug	22-Aug	ongoing		
yes	all	Key Account Manager	consider offering a hose inspector, to improve machine reliability and support business	15-Dec	08-Nov	ongoing	10 Aug: waiting for input from KAM	8 December: Will be discussed during the visit on site this weekend
yes	777E KYD	Country Parts Manager	confirm RDSL orders placed	01-Jun	31-May	ongoing	21 Jun: no specific stocking for the 777E KYD prefix was done, reminder send on actions taken since	
yes	all	Country Technical Manager	it was identified during the training for the hose team that there is no maintenance of the equipment, creating a risk of producing hoses out of specification. Paul visit planning	01-Aug	28-Mar	ongoing	10 Aug: SPM to confirm with Parts Manager	15 April: Several options are currently being discussed between dealership and the central team. I do not want to speak on their behalf, they will come back to you once the investments decisions are made, the sooner the better.)
no	6020B	KAM/ SPM	Customer has concerns about productivity of 6020B older units next steps are: 1. TA2 inspection by Dealer Expert technician 2. Feedback to GM on the production study led by the dealer	15-Sep		ongoing		
no	MD6240	PSSR/ service director	source track frame to replace cracked/ propose repair o do we really need to replace U/C with non-Cat o can the track frame be repaired (dealer)	01-Sep	01-Jun	ongoing		
yes	all	PSSR/ SPM	General monitor A/C related downtime trend o make a list of maintenance operations from OMM and builder and share with customer PSSR (done) - complete o prepare training offer to customer SPM/ PSSR o ensure stock for A/C parts is available on VMI Dealer Site Manager o offer technician for A/C from dealer, include tooling	01-Jun	01-May	ongoing	10 Aug: trainer will arrive in August	8 Feb: training planned June
no	MD6250	Site Manager	o evaluate condition of coolers/ radiators of MD6250 and undertand frequency of cleaning	15-Apr	06-Apr	ongoing	10 May: customer does not clean radiators frequently enough according to dealer, however during SPM visit all radiators inspected were clean	
yes	6040FS	Cat Tech Rep/ Dealer Technical Director	track issues o settle repair with the customer o order tracks for stock o investigate issue	15-Apr	15-Mar	ongoing	10 Aug: Christian to provide feedback on parts ordered and position	10 May: pins ordered replace pins on 6040FS CTS tracks order track for FS02
yes	D10 downtime on UC	PSSR	review CTS/ liaise with GET expert/undertand root cause of downtime/ offer service to the customer	10-Sep	10-Aug	ongoing		
no	all	SPM/ Dealer Site Manager	Elimination of Live Work, share all available instructions SPM/ consider supporting implementation Dealer Site Manager	20-Dec	09-Dec	complete		
no	789D	TC	AFA on TC on HT?? Failed in November	20-Dec	09-Dec	complete		
no	6040FS	Site Manager	share report for block valve issue, which caused the pump failures	12-Nov	08-Nov	complete		
yes	6040FS	Site Manager	CTS on 6040FS	10-Mar	10-Feb	complete		
yes	6020B	Site Manager	service kit for brakes has the wrong parts in it, 6 kits received with this issue, make a DSN report	10-Dec	08-Dec	complete	10 January: order was correctly filled	

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2.8 Roles and Responsibilities

It's important to assign roles such that each person is clear what their contributions and deliverables are as part of the continuous improvement process. Some examples are noted below:

Gather initial data	Planner/ reliability analyst
Identify top 5 / top 10 issues by category	Planner/ Reliability analyst
Perform Root Cause Analysis – identify opportunities in maintenance, operations, parts logistics, quality, etc.	Planner/ Reliability analyst/ Condition Monitoring/ technical experts
Define improvement actions	Planner/ Reliability analyst/ Condition Monitoring/ technical experts
Regular meeting/ follow-up	Maintenance Manager/ Planner/ reliability analyst / CI/ Dealer Site Manager/ PSSR/ SPM/ tech rep

Note: If site has a Caterpillar or Dealer SPM assigned, the SPM can overview process and drive execution of deliverables, especially opportunities identified.

3.0 Implementation Steps

The first step is to get agreement or alignment with the customer, Dealer and Caterpillar personnel about the objectives of the exercise, roles & responsibilities, and resources. Customers often have apprehension about sharing internal information so developing a trusting relationship is necessary prior to this stage. Agreement and commitment of all parties (customer, dealer, Caterpillar) at a management level helps get traction and start the process in the right way.

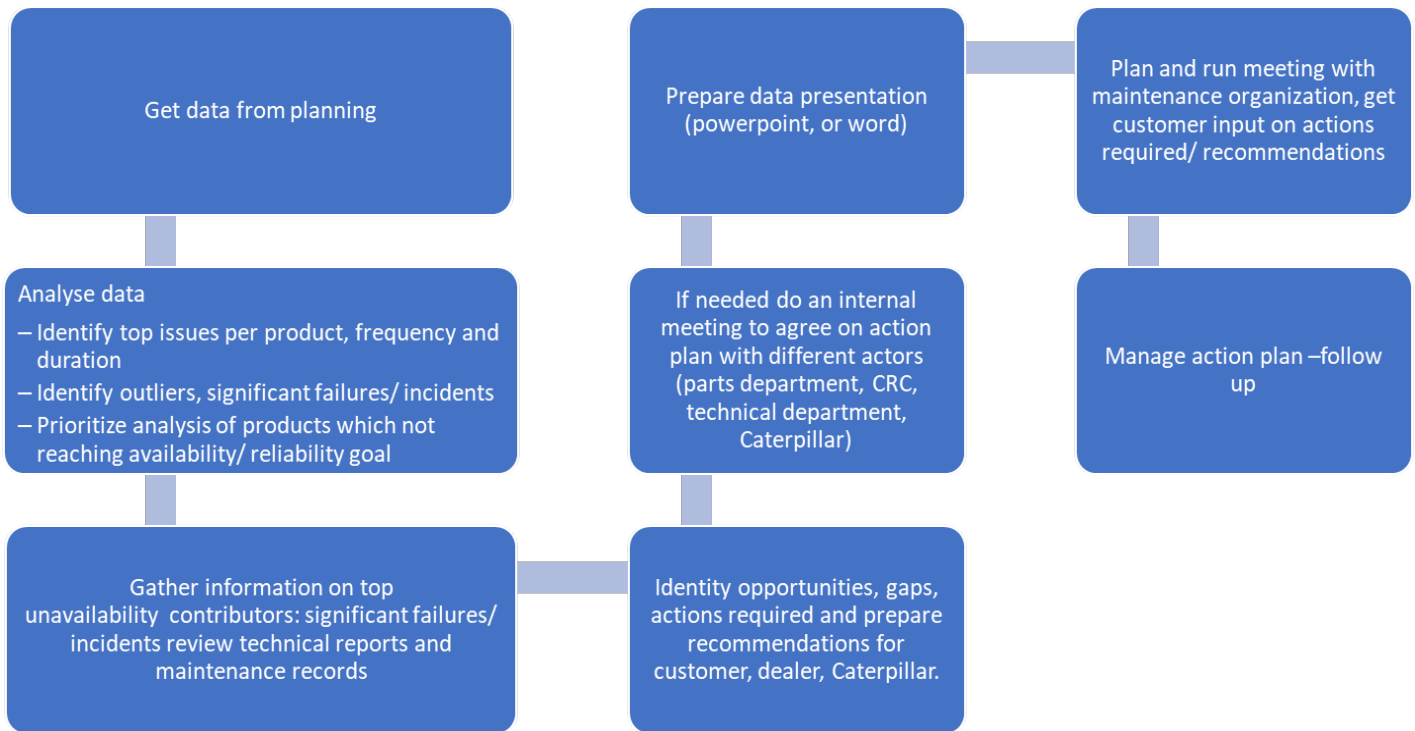
An important reminder is that no data is perfectly accurate. Therefore, it's suggested not to delay the beginning of the analysis due to quality of initial source data. Valuable observations, conclusions and recommendations are possible even with poor-quality information. Initial wins will boost morale and encourage the maintenance organization to improve data accuracy and improve the quality of recommendations and action plans that can be implemented.

Attached is a list of recommended actions, performed on a monthly basis:

1. Get data from planning department.
2. Analyze data:
 - Identify top issues per product, frequency and duration.
 - Identify outliers, significant failures/ incidents.
 - Prioritize analysis of products which not reaching availability/ reliability goal.
3. Gather information about top unavailability contributors: significant failures/ incidents review technical reports and maintenance records.
4. Identify opportunities, gaps, actions required and prepare recommendations for customer, dealer, Caterpillar.
5. If needed, conduct an internal meeting to agree on action plan with different areas (parts department, CRC, technical department, Caterpillar).

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6. Prepare data presentation (PowerPoint or Word).
7. Plan and conduct a meeting with the maintenance organization and get customer input on actions required / recommendations.
8. Manage action plan and follow up.



4.0 Benefits

- Improve availability by optimizing resource utilization in areas with highest opportunity.
- Dealer, customer, Caterpillar work towards the same goals and align priorities.
- Improved relationship and trust with the customer.
- Data based approach allows efficient use of resource on the most urgent issues.

5.0 Resources Required

- Agreement with customer to share maintenance data.
- Dispatch and/ or maintenance system to document downtime.
- Process owner can be Site Manager, Site Performance Manager or Reliability Engineer.

6.0 Supporting Attachments / References

- Sample report
- Caterpillar Mining Equipment Management (MEM) Performance Metrics [Caterpillar Media Publications: PEBJ0190 (English), PSBJ90190 (Spanish), PRBJ0190 (Russian), PPBJ0190 (Portuguese)]

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- Caterpillar Service Management Control System (SMCS) [SMCS website](#)

7.0 Related Best Practices

- BP 1311-1.09-1326: Performance Evaluation (Strategic)
- BP 1.03-210611-1: Strategic Plan to Recover the Performance of a Mining Equipment Fleet
- BP 1007-2.0-1000: Continuous Improvement (Strategic)

8.0 Acknowledgements

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